

MATH 8090: ARMA Models: Properties, Identification, and Estimation

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Checking Stationarity and Causality

Consider the following AR(4) process

$$\eta_t = 2.7607\eta_{t-1} - 3.8106\eta_{t-2} + 2.6535\eta_{t-3} - 0.9238\eta_{t-4} + Z_t,$$

the AR characteristic polynomial is

$$\phi(z) = 1 - 2.7607z + 3.8106z^2 - 2.6535z^3 + 0.9238z^4$$

```
# Calculate the roots of the polynomial and store them in xi
xi <- polyroot(c(1, -2.7607, 3.8106, -2.6535, 0.9238))
xi
```

```
## [1] 0.6499635+0.7859373i 0.7862240+0.6500408i 0.6499635-0.7859373i
## [4] 0.7862240-0.6500408i
```

```
# Calculate the moduli of the roots
Mod(xi)
```

```
## [1] 1.019877 1.020148 1.019877 1.020148
```

PACF

```
# =====
# Example 1: AR(2) process
#  $\eta_t = 0.5 \eta_{t-1} + 0.25 \eta_{t-2} + z_t$ 
# =====

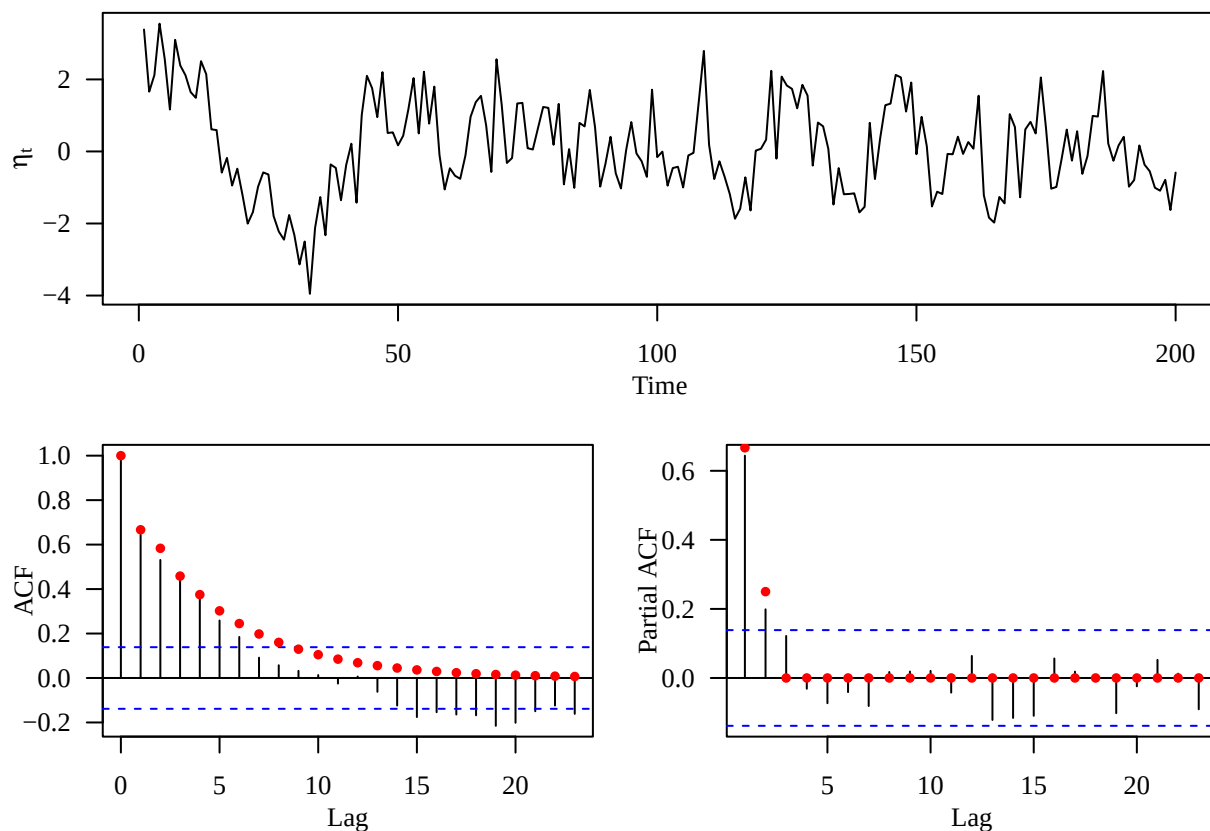
# Simulate 200 observations from an AR(2) process
ar2_1 <- arima.sim(n = 200, list(ar = c(0.5, 0.25)))

# Plot the simulated series, sample ACF, and sample PACF
par(las = 1, mgp = c(2, 1, 0), mar = c(3.6, 3.6, 0.8, 0.6), family = "serif")
layout(matrix(c(1, 1, 2, 3), 2, 2, byrow = TRUE))

ts.plot(ar2_1, ylab = expression(eta[t]))

# Sample ACF with the theoretical ACF overlaid in red
acf(ar2_1)
acf_true <- ARMAacf(ar = c(0.5, 0.25), lag.max = 23)
points(0:23, acf_true, pch = 16, cex = 0.8, col = "red")

# Sample PACF with the theoretical PACF overlaid in red
pacf(ar2_1)
pacf_true <- ARMAacf(ar = c(0.5, 0.25), lag.max = 23, pacf = TRUE)
points(1:23, pacf_true, pch = 16, cex = 0.8, col = "red")
```



```
# =====
# Example 2: MA(1) process
```

```

# eta_t = z_t + z_{t-1}
# =====

# Simulate 200 observations from an MA(1) process
ma1 <- arima.sim(n = 200, list(ma = c(1)))

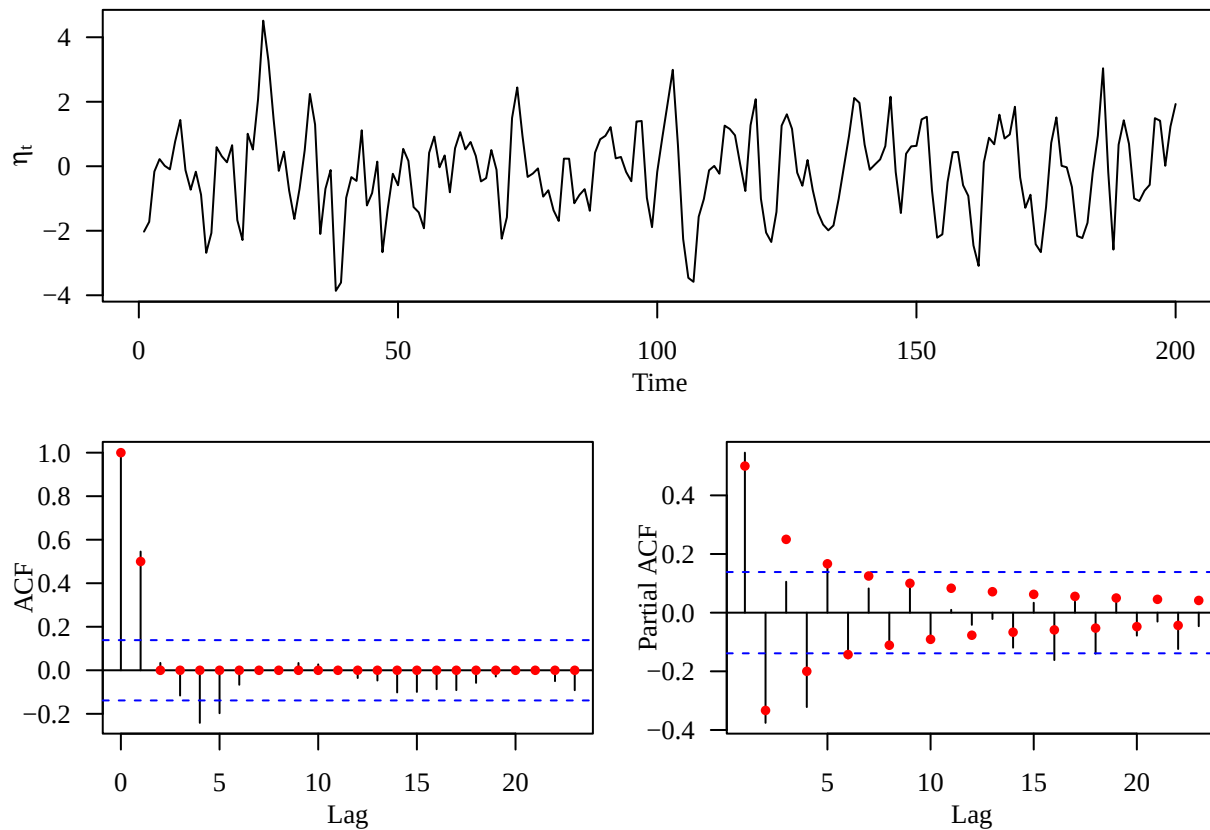
# Plot the simulated series, sample ACF, and sample PACF
par(las = 1, mgp = c(2, 1, 0), mar = c(3.6, 3.6, 0.8, 0.6), family = "serif")
layout(matrix(c(1, 1, 2, 3), 2, 2, byrow = TRUE))

ts.plot(ma1, ylab = expression(eta[t]))

# Sample ACF with the theoretical ACF overlaid in red
acf(ma1)
acf_true <- ARMAacf(ma = c(1), lag.max = 23)
points(0:23, acf_true, col = "red", cex = 0.8, pch = 16)

# Sample PACF with the theoretical PACF overlaid in red
pacf(ma1)
pacf_true <- ARMAacf(ma = c(1), lag.max = 23, pacf = TRUE)
points(1:23, pacf_true, col = "red", cex = 0.8, pch = 16)

```



```

# =====
# Example 3: ARMA(2,1) process
# eta_t = 0.5 eta_{t-1} + 0.25 eta_{t-2}
#       + z_t + z_{t-1}

```

```
# =====

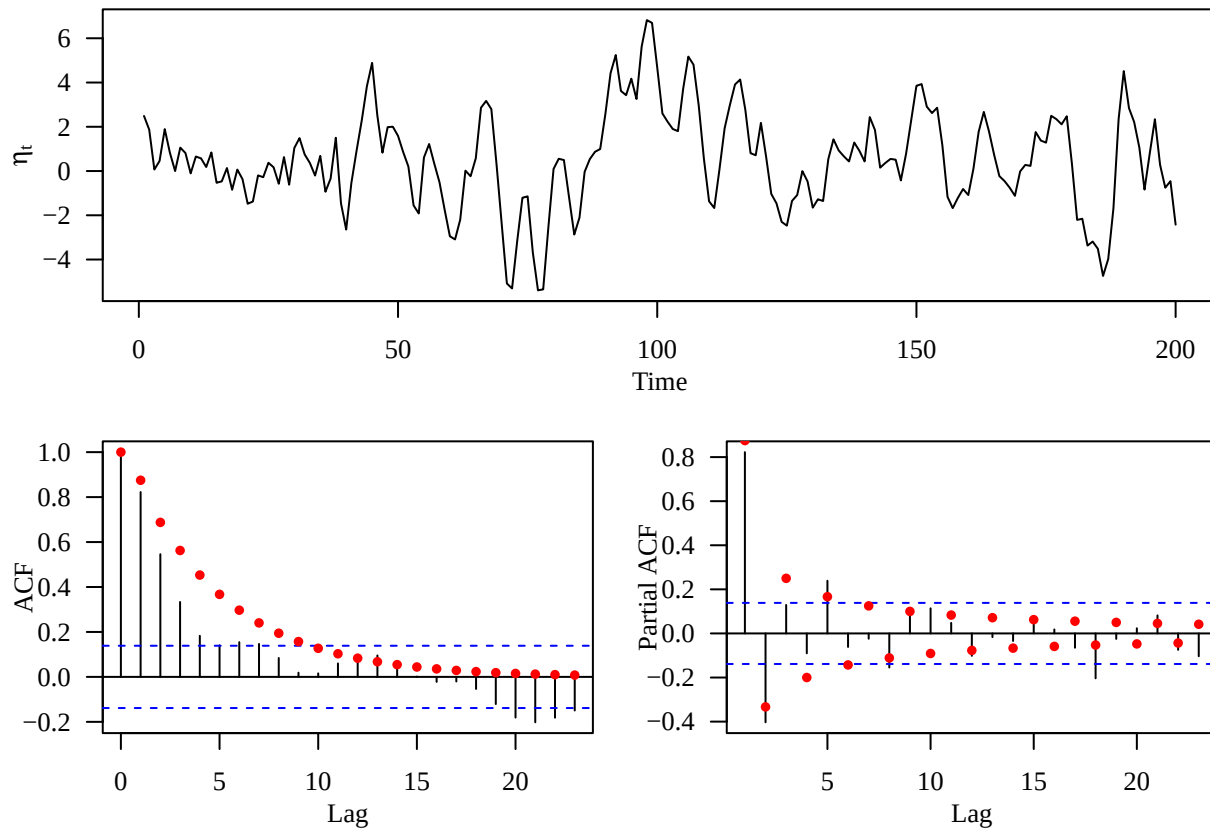
# Simulate 200 observations from an ARMA(2,1) process
arma <- arima.sim(n = 200, list(ar = c(0.5, 0.25), ma = c(1)))

# Plot the simulated series, sample ACF, and sample PACF
par(las = 1, mgp = c(2, 1, 0), mar = c(3.6, 3.6, 0.8, 0.6), family = "serif")
layout(matrix(c(1, 1, 2, 3), 2, 2, byrow = TRUE))

ts.plot(arma, ylab = expression(eta[t]))

# Sample ACF with the theoretical ACF overlaid in red
acf(arma)
acf_true <- ARMAacf(ar = c(0.5, 0.25), ma = c(1), lag.max = 23)
points(0:23, acf_true, col = "red", cex = 0.8, pch = 16)

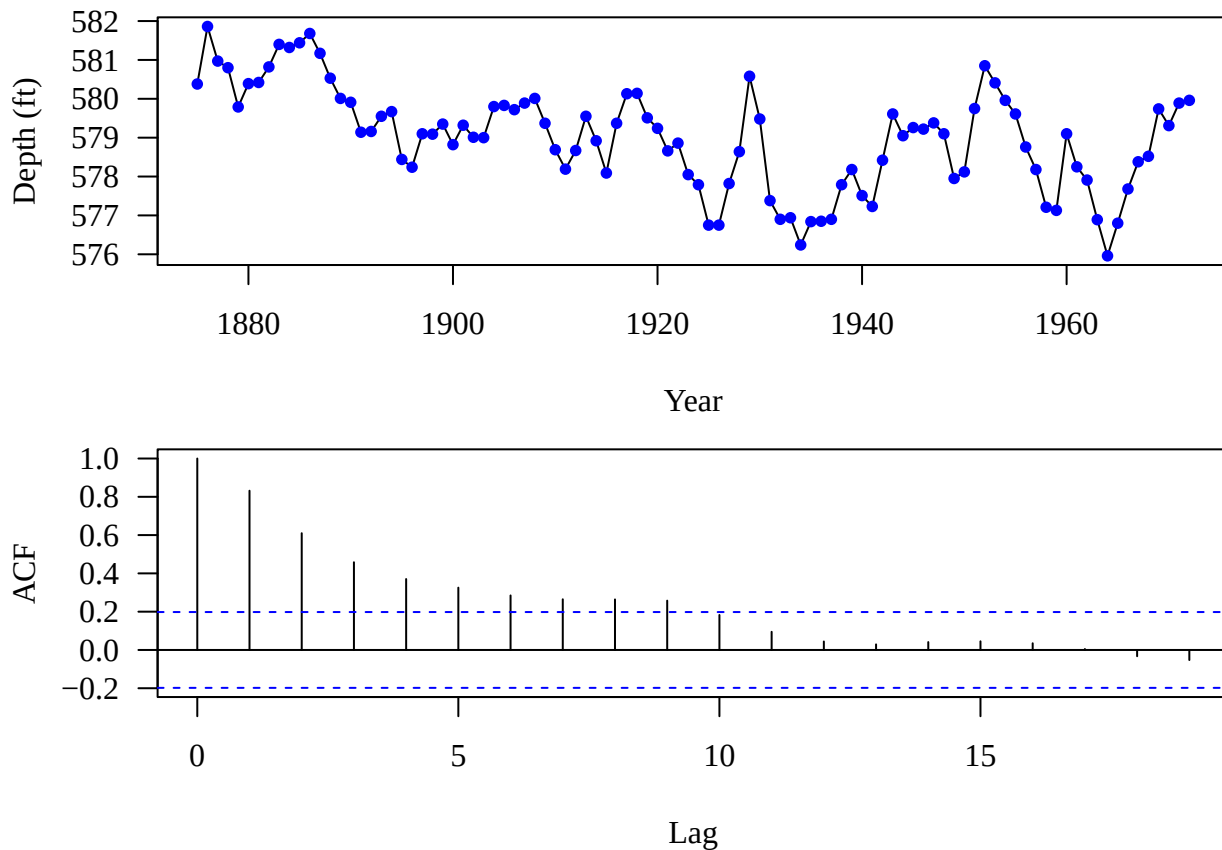
# Sample PACF with the theoretical PACF overlaid in red
pacf(arma)
pacf_true <- ARMAacf(ar = c(0.5, 0.25), ma = c(1),
                    lag.max = 23, pacf = TRUE)
points(1:23, pacf_true, col = "red", cex = 0.8, pch = 16)
```



Lake Huron Example

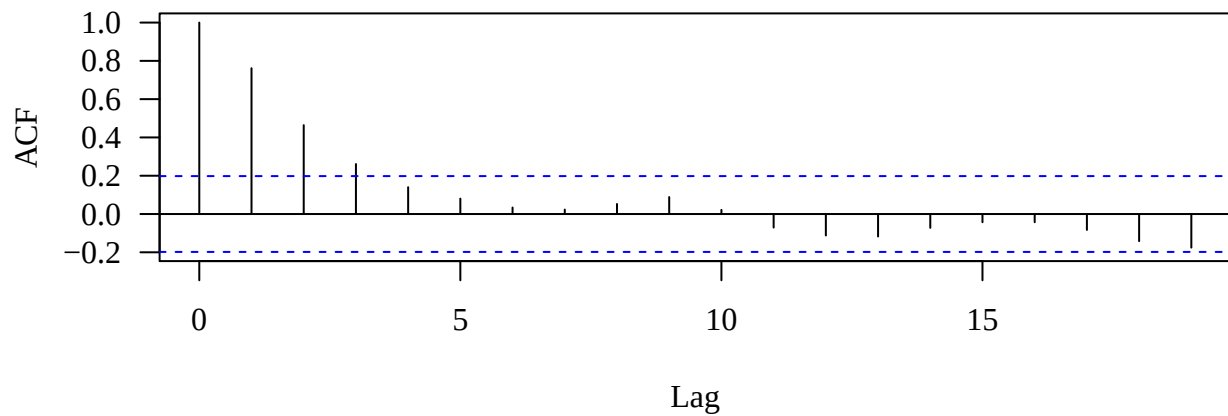
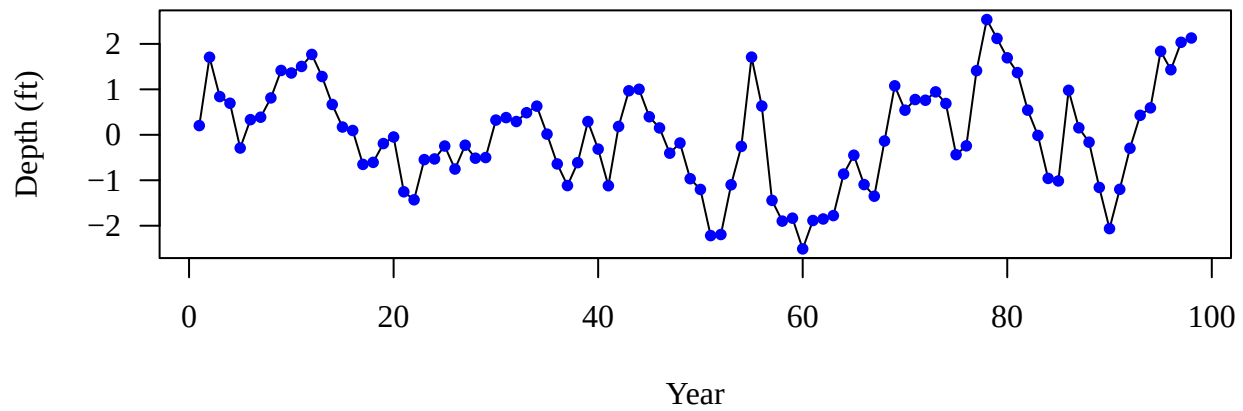
```
# Load the Lake Huron annual water-level data
data(LakeHuron)

# Plot the original time series and its sample ACF
par(las = 1, mfrow = c(2, 1), mar = c(4, 4, 0.8, 0.6), family = "serif")
plot(LakeHuron, ylab = "Depth (ft)", xlab = "Year")
points(LakeHuron, cex = 0.8, col = "blue", pch = 16)
acf(LakeHuron)
```



```
# Remove the linear trend from the Lake Huron series
yr <- 1875:1972
lm <- lm(LakeHuron ~ yr)

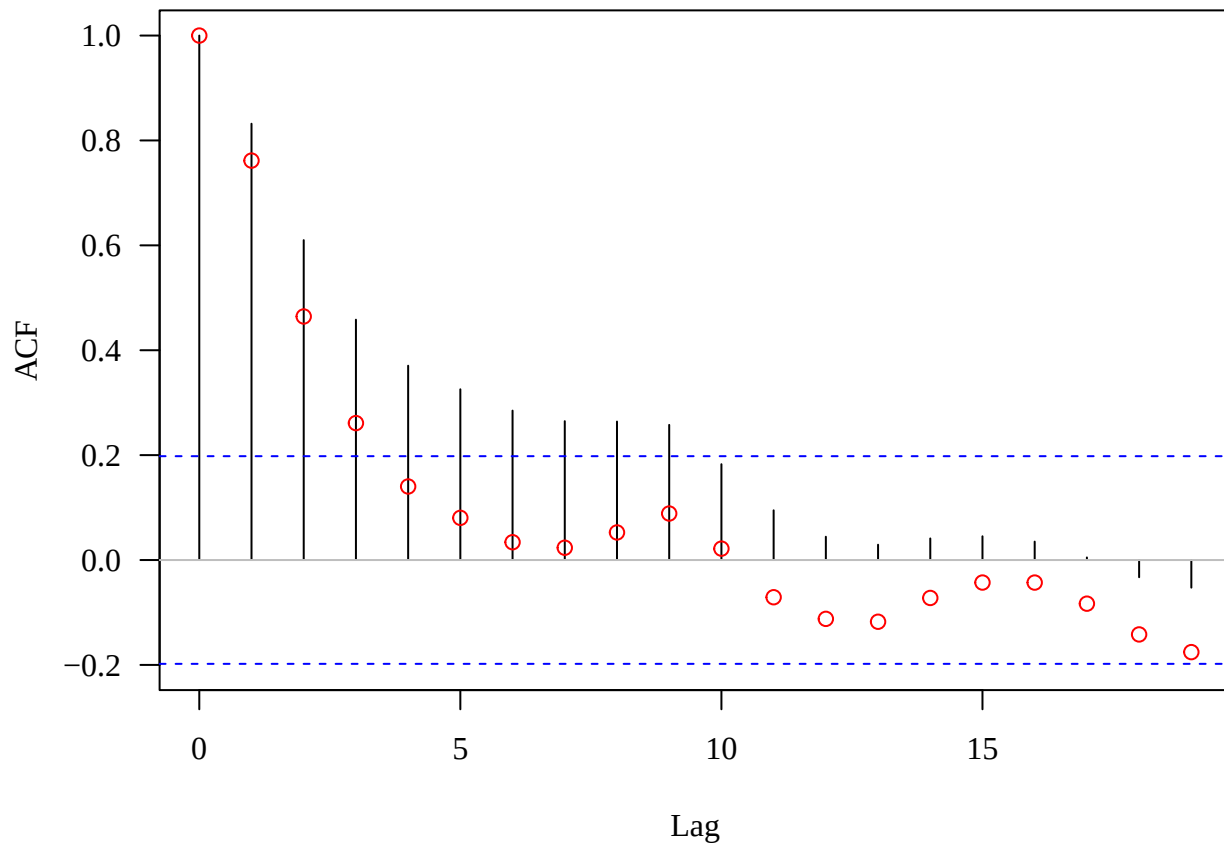
# Plot the detrended series (residuals) and its sample ACF
plot(lm$residuals, ylab = "Depth (ft)", xlab = "Year", type = "l")
points(lm$residuals, cex = 0.8, col = "blue", pch = 16)
acf(lm$residuals)
```



```
# Compare the sample ACF before and after removing the linear trend
par(mfrow = c(1, 1), las = 1)
plot(0:19, acf(LakeHuron, plot = FALSE)$acf,
     type = "h", xlab = "Lag", ylab = "ACF", ylim = c(-0.2, 1))
abline(h = 0, col = "gray")

# Add approximate 95% white-noise bounds
abline(h = c(-1, 1) * qnorm(0.975) / sqrt(length(LakeHuron)),
       col = "blue", lty = 2)

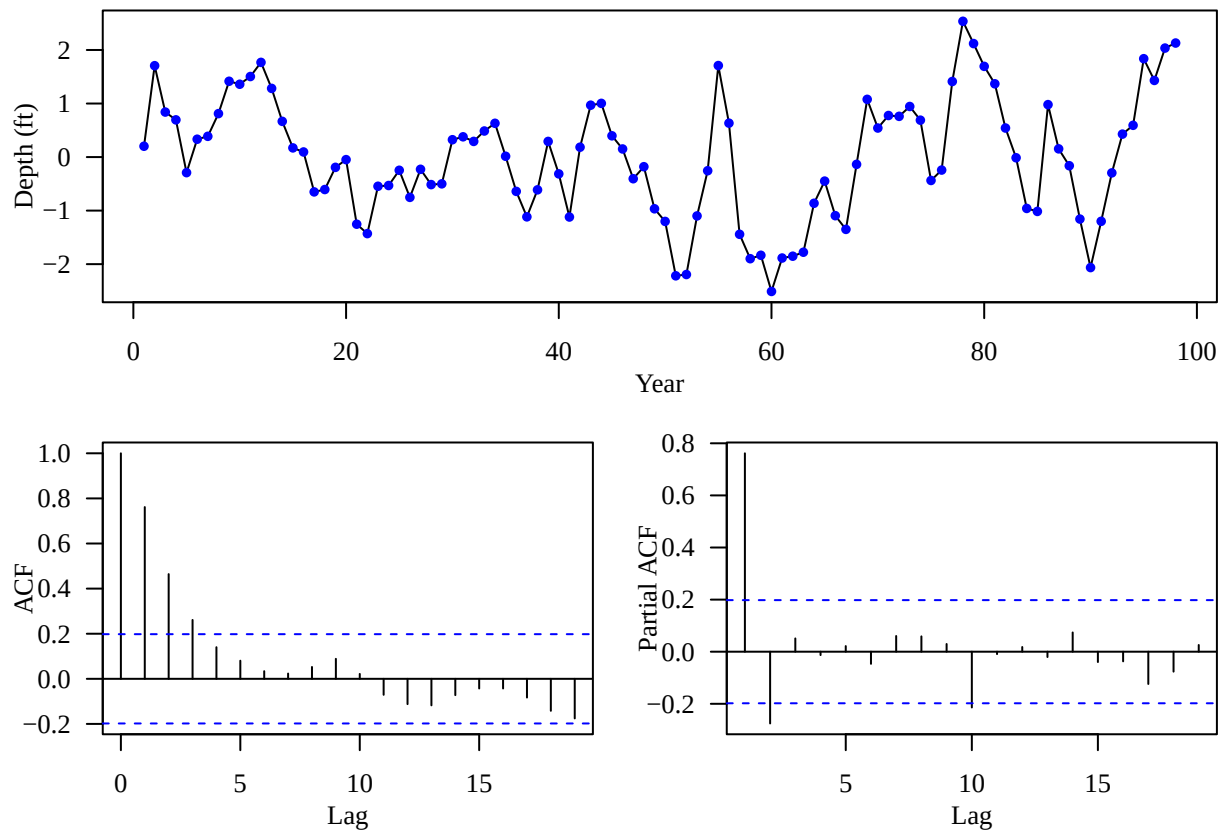
# Overlay the sample ACF of the detrended series in red
acf_detrend <- acf(lm$residuals, plot = FALSE)$acf
points(0:19, acf_detrend, col = "red")
```



```
# Examine the temporal dependence remaining after detrending
par(las = 1, mgp = c(2, 1, 0), mar = c(3.6, 3.6, 0.8, 0.6))
layout(matrix(c(1, 1, 2, 3), 2, 2, byrow = TRUE))

# Detrended Lake Huron series
plot(lm$residuals, ylab = "Depth (ft)", xlab = "Year", type = "l")
points(lm$residuals, cex = 0.8, col = "blue", pch = 16)

# Sample ACF and PACF of the detrended series
acf(lm$residuals)
pacf(lm$residuals)
```



An example of Yule-Walker estimate

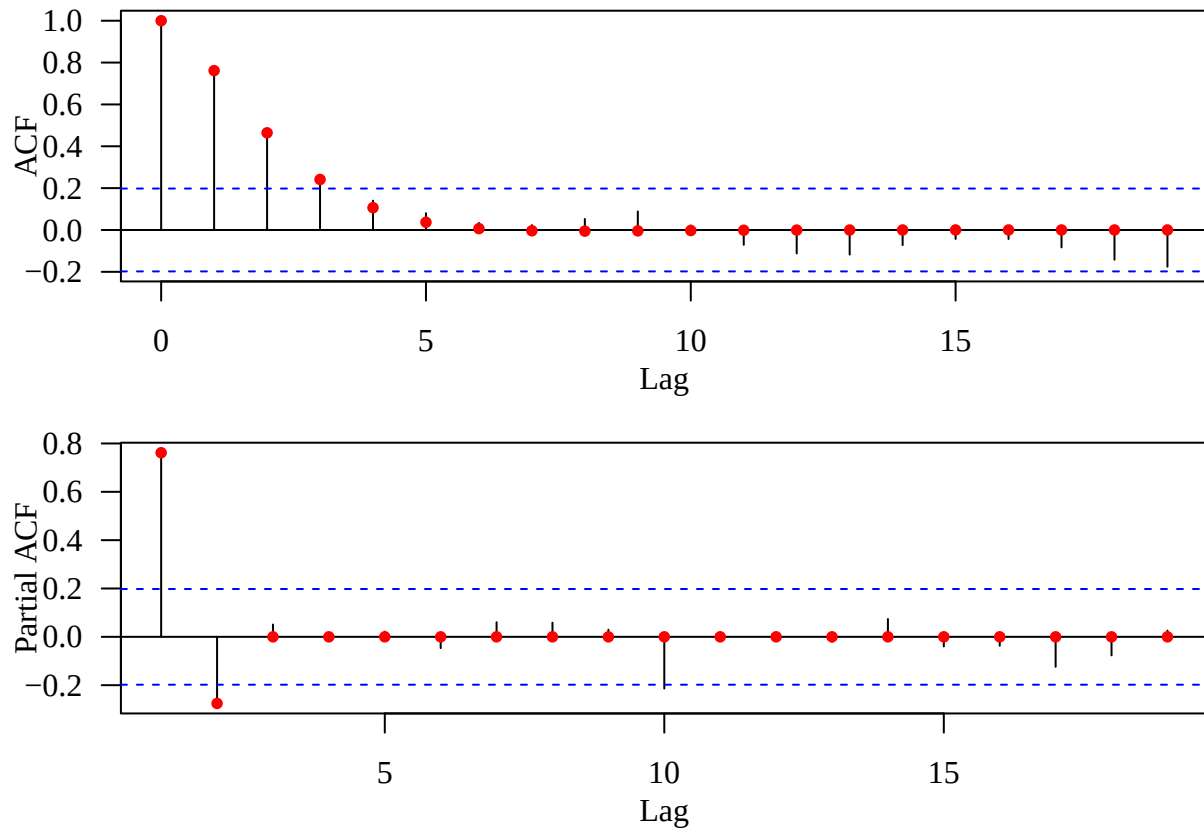
Let's fit an AR(2) to the detrended lake huron series

```
# Fit an AR(2) model to the detrended series using Yule-Walker estimation
YW_est <- ar(lm$residuals, aic = FALSE, order.max = 2, method = "yw")

# Compare the sample ACF/PACF with those implied by the fitted AR(2) model
par(las = 1, mgp = c(2, 1, 0), mar = c(3.6, 3.6, 0.6, 0.6),
    mfrow = c(2, 1), family = "serif")

# Sample ACF with the fitted AR(2) theoretical ACF overlaid in red
acf(lm$residuals)
acf_YWest <- ARMAacf(ar = YW_est$ar, lag.max = 23)
points(0:23, acf_YWest, col = "red", pch = 16, cex = 0.8)

# Sample PACF with the fitted AR(2) theoretical PACF overlaid in red
pacf(lm$residuals)
pacf_YWest <- ARMAacf(ar = YW_est$ar, lag.max = 23, pacf = TRUE)
points(1:23, pacf_YWest, col = "red", pch = 16, cex = 0.8)
```



MLE

```
# Fit an AR(2) model to the detrended series using maximum likelihood
(MLE_est1 <- arima(lm$residuals, order = c(2, 0, 0), include.mean = FALSE))
```

```
##
## Call:
## arima(x = lm$residuals, order = c(2, 0, 0), include.mean = FALSE)
##
## Coefficients:
##          ar1      ar2
##      1.0050  -0.2925
## s.e.  0.0976   0.1002
##
## sigma^2 estimated as 0.4572:  log likelihood = -101.26,  aic = 208.51
```

```
# Fit an AR(1) model to the detrended series for comparison
(MLE_est2 <- arima(lm$residuals, order = c(1, 0, 0), include.mean = FALSE))
```

```
##
## Call:
## arima(x = lm$residuals, order = c(1, 0, 0), include.mean = FALSE)
##
## Coefficients:
```

```
##          ar1
##          0.7826
## s.e.    0.0635
##
## sigma^2 estimated as 0.4975:  log likelihood = -105.32,  aic = 214.65

# Fit an ARMA(2,1) model to the detrended series for comparison
(MLE_est3 <- arima(lm$residuals, order = c(2, 0, 1), include.mean = FALSE))

##
## Call:
## arima(x = lm$residuals, order = c(2, 0, 1), include.mean = FALSE)
##
## Coefficients:
##          ar1          ar2          ma1
##          0.8381 -0.1631  0.1842
## s.e.    0.3178   0.2618  0.3179
##
## sigma^2 estimated as 0.4556:  log likelihood = -101.09,  aic = 210.19

# Jointly estimate the linear trend and AR(2) dependence
# using the original Lake Huron series
(MLE_est4 <- arima(LakeHuron, order = c(2, 0, 0), xreg = yr))

##
## Call:
## arima(x = LakeHuron, order = c(2, 0, 0), xreg = yr)
##
## Coefficients:
##          ar1          ar2  intercept          yr
##          1.0048 -0.2913   620.5115  -0.0216
## s.e.    0.0976   0.1004    15.5771   0.0081
##
## sigma^2 estimated as 0.4566:  log likelihood = -101.2,  aic = 212.4
```

Diagnostic

```
# Check whether any significant temporal dependence remains in the residuals
# from the Yule-Walker AR(2) fit
Box.test(YW_est$resid[-(1:2)], lag = 10, type = "Ljung-Box", fitdf = 2)

##
## Box-Ljung test
##
## data:  YW_est$resid[-(1:2)]
## X-squared = 4.2709, df = 8, p-value = 0.8319

# Check whether any significant temporal dependence remains in the residuals
# from the jointly fitted trend + AR(2) model
Box.test(MLE_est4$residuals, lag = 10, type = "Ljung-Box", fitdf = 2)
```

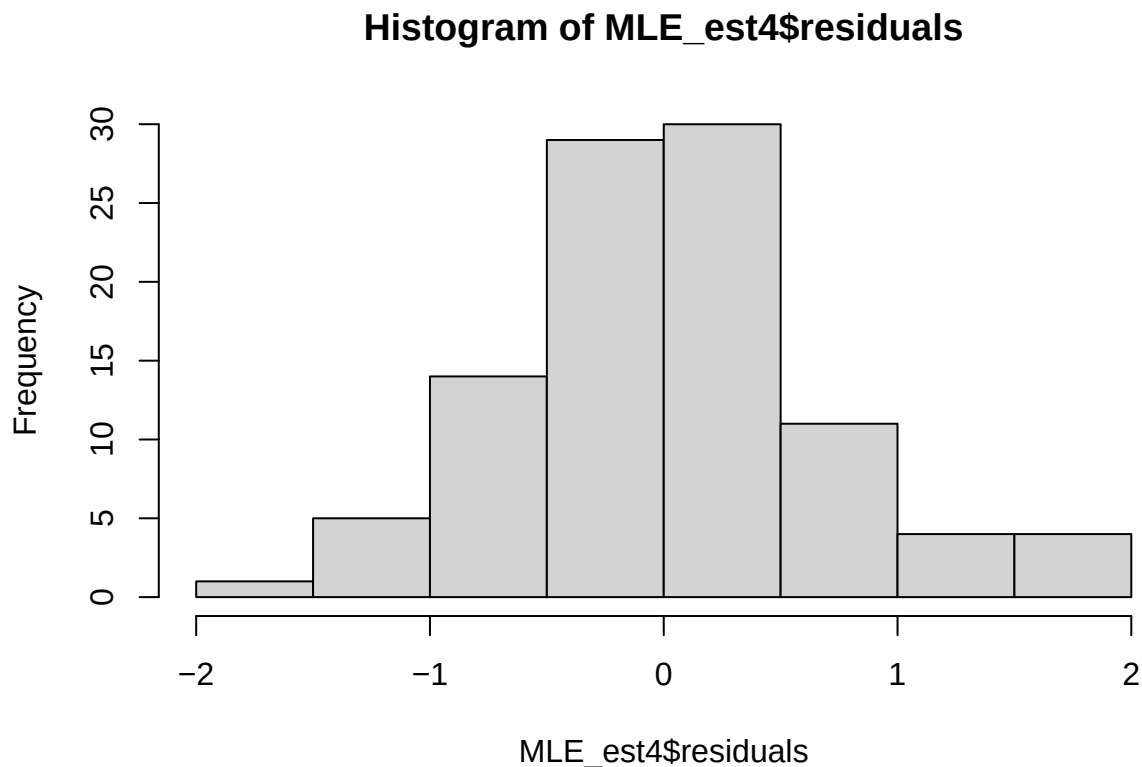
```
##
## Box-Ljung test
##
## data: MLE_est4$residuals
## X-squared = 3.9283, df = 8, p-value = 0.8635

# Fisher-Gallagher weighted portmanteau test
# Gives greater weight to residual autocorrelations at smaller lags
library(WeightedPortTest)

Weighted.Box.test(MLE_est4$residuals, lag = 10, type = "Ljung-Box", fitdf = 2)

##
## Weighted Ljung-Box test (Gamma Approximation)
##
## data: MLE_est4$residuals
## Weighted X-squared on Residuals for fitted ARMA process = 1.115, Shape
## = 8.17568, Scale = 0.67273, p-value = 0.9998

# Examine the distribution of the residuals using a histogram
hist(MLE_est4$residuals)
```



```
# Assess the normality of the residuals using a normal Q-Q plot
par(las = 1, mgp = c(2, 0.7, 0), mar = c(3.6, 3.6, 0.6, 0.6), family = "serif")
qqnorm(MLE_est4$residuals, col = "blue", cex = 0.8)
qqline(MLE_est4$residuals)
```

